Assessment 02 - Classical cryptographic schemes

Instructions

The assessment must be developed individually and submitted via Blackboard by **09/10/2026**. The submission should be a *zip* file containing all developed code and a short PDF report describing the code and discussing the results achieved.

Activities

Q1: The following ciphertext was generated using a monoalphabetic cipher using an unknown `key` and permutation of the alphabet `ABCDEFGHIJKLMNOPQRSTUVWXY`. You know that the *plaintext* is written in Portuguese. Thus, apply a cryptanalysis based on the frequency table in this link to discover the `permutation`, `key`, and `plaintext`.

```
JTJZHJTIKTCJZKITJTTEYJGJPKTFVIPJKBEPIYSJGDZJEJGVTESJYJDKZHJZITYVYBJPIJYSITYJUILJPKTDJTTJZJHJEYPJJGIHPJSJDZKCJY
JIHDIZELKTILVIZZJTITQKZBJPKTHJETPKFVIDZKHISEJJQKZBJMVHJYJIIYSZILIYSIZIHKSJIPEQEBJZJHYKUKZIEYKFVISJYSKTVCGEHJZJ
HISJHCIHJTHIHKZEJTLGKZEKTJTPJFVIGITZIETFVIQKZJHPEGJSJYPKJQIKEHDIZEKIJTSIZZJTUEBEKTJTPIJQZEBJIPIJTEJJYPJZJHPIUJ
TSJYPKIJFVIGITFVIDKZKCZJTUJGIZKTJTTIUJKPJGIEPJHKZSIGECIZSJYPKBJYSJYPKITDJGMJZIEDKZSKPJDJZSITIJSJYSKHIJAVPJZKIY
LIYMKIJZSIBITTIHPKTJCEKLZILKIPKSZKEJYKJTYJUILJBKITLZJYPITFVIQERIZJHBJGITIPIJGIWJYPZKIPISZJAJYKJQJHJPJTUESKZEJT
FVISEUIZJHFVIIVBJYSKKDIESKEGVTSZIGVTESJYKJFVIHYIDSVYKIHJZSIKCIPIBIZJHBITTISVPKKFVIJHVTJJYSELVJBJYSJFVIKVSZKUJG
KZHJETJGSKTIJGIUJYSJIUKTSJLEPITHEYMJTDKETBZEJPKSIYPITIHHEHVHYKUKIYLIYMKJZPIYSITITIHDZIIHUIZTKMVHEGPIBIGICZJPKQ
KEPIHEHUKTTKZEKJGILZIHIYSIPJEHIJLKZJVHTKHJGSKITVCGEHJPKVHITSEGKLZJYPEGKFVKIBKZZIYSIDKZFVIPIUKTTJTJLVJTQICKKZPI
YIFVIYJKSIYMJHEYUIAJJTPIMEDKIZIYIPJEHIVHJQVZEJLZJYPIITKYKZKTJIYJKPIJLZITSIJUIYJKVQZJVSJZVPJHJTPISVCJBJYKZJICIG
EBKTJFVIKDIESKJBIYPIIJBKZJKLITSKHVPJPJEHIELVJGBJYSKJKTQIESKTPJQJHKTJLIYSIUKTTJFVIJHJZSISJYSKJAVPJFVITIITDJGMII
TIBJYSIYKVYEUIZTKTISJKTVCGEHIDZIBKBJCIIHUIZTKIUKTKCIHYJTBEPJTILVZJYBJPJGVTESJYJJYSELVJGECIZPJPIIYJKHIYKTBIZSET
TEHJITDIZJYBJPIJVHIYSKPJDIFVIYJBZETSJYPJPIUKTKYKUKSIHKZPJHJVZJGJYBJHJZJUEGMJQJSJGPJYKTTJEPJPIPJPJJKHVYPKDKZPIV
TFVISKPKKHJYPIDJZJPKHVYPKJPIVTPJZDJZSILZJYPIUKTSIYZKIYKUKZJHKQGKZITBIYSIPIVHJJZUKZIPIBZETSKHJETJHJPJFVIYIYMVHJ
YJTBEPJYKKBEPIYSIBITJZIJKVBZETSEJYETTEHJBMJHJPJUIPIKYKUKTTKITBVPKFVIDZITIYSIUKTJHKTSZJJUESKZEJAJDJTTJPJYJFVJGU
KTPIVDKZJZHJTIPIEWKVJTFVIIGIDJZJTEYJBZVRSKHKVUKTDKPIZKTKZIEBVAKJGSKEHDIZEKKTKGGKLKIHYJTBIYPKUIDZEHIEZKUIKSJHCI
HYKHIEKPKMIHETQIZEKIFVJYPKPITBIKPIEWJPIZZJPIEZKUKTFVIITDIZJHKTAVLKIUESVDIZEKPKSKZDIETHJIGESJBJUJGIEZKPKSVZBKKZ
EIYSJGIPKLIYSEKFVIEYPJCICIKGEBKZPKTJYSKZEKEYBGEYJEDKZVHDKVBKJHJAITSJPIFVIYITTISIYZKLITSKUKTBKYSIHDGKFVIAJTIHKT
SZJFVJGYJEYSIEZJEPJPIFVJYPKTVCEYPKEZIETJKISIZYKSIHDGKKTKGMKTPJZIJGCIYELYEPJPIDKYPIYKBMJKUIZIETVHYKUKIWIHDGKPIJ
HKZPKTDJSZEKTQIESKTUJGIZKTKTIHUIZTKTPEUVGLJPKYVHIZKTKTUIZIETJHKZPJDJSZEJYJKHKUEPKPIDZIHEKUEGHJTJGSKIFVJTIISIZY
KFVIYJKIDZIHEKUEGTIZBKYMIBEPKDKZVHDZILJKPKYEYMKHIVDJSIZYKKVUEUIZIETKYKHIIYLZJYPIBEPKPJFVIGITPIFVIHTKETTIYMKZTV
DIZYKIAVGLJZIETFVJGIHJETIWBIGIYSITITIZPKHVYPKZIETIPISEGLIYSI
```

To simplify the process, all special characters and spaces were removed. Also, consider only upper-case text.

Q2: Attacking the Vigenère cipher is considerably more challenging. The problem is usually approached by considering two sub-problems: 1) determining likely key sizes; and 2) attacking individually each of the “slices” of the *ciphertext* encrypted by each character of the key. The first of these problems already requires considerable sophistication, so we will focus on the second: attacking the various Caesar ciphers used.

Write the program vigenere*attack.py that performs an “attack” on the Vigenère cipher. The program should receive the following arguments:*
• *the key size;*
• *the file path to the ciphertext under analysis;*
• *a non-empty sequence of words that may be found in the _plaintext* (it is assumed that at least one of the provided words occurs in the plaintext).
As output, the program should produce:
• an empty response if it cannot find the key used to encrypt the *ciphertext*;
• two lines in case of success:
• the first line containing the key used in the cipher;
• the second line containing the complete plaintext.

```
$ python3 vigenere_attack.py 3 file_path.txt PAPO PRAIA POR
ASARMASEOSBAROESASSINALADOSQUEDAOCIDENTALPRAIALUSITANA

#Assume the content of file_path.txt is
"PGRGARHSFHPRGCVHOJHWEPZRSCJFIVSOFRWUTBKPZGGOZPZLHWKPBR"
```

Use your program to break the *ciphertext* within the file *Q2_ciphertext.txt*, assuming the used key size is 4, the *plaintext* is written in Portuguese, and at least one of the following words appears in the *plaintext*: DIREITO, FANTASIA, LIBERDADE, MUNDO, QUE, and INTERVENCAO.

To simplify the process, all special characters and spaces were removed. Also, consider only upper-case text.